

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID *****

Abstract

001 Evaluating AI/ML systems responsibly requires prac-
002 tices for testing, stress-testing, auditing, comparing,
003 and interpreting specific capabilities—not only aggre-
004 gate benchmark scores. This paper proposes such a
005 practice for spatial reasoning in vision–language mod-
006 els (VLMs): a diagnostic audit of object-relative orien-
007 tation, a capability that aggregate accuracy can hide.
008 On Visual Spatial Reasoning (VSR), moving from a
009 compact 2.2B model to a larger 7B VLM substan-
010 tially improves overall, depth, horizontal, and con-
011 tainment accuracy, but orientation changes only mod-
012 estly; a frontier-scale 2026 model (MiMo-V2.5, 310B
013 total / 15B active) reproduces the pattern (79.5%
014 overall, 65.7% orientation). Except for structured
015 prompting, which degrades performance, trained con-
016 ditions on both models and the frontier-scale zero-
017 shot model remain in a narrow 62–66% orientation
018 band. We test structured prompting, general and tar-
019 geted LoRA, hard negatives, visual/projector adapta-
020 tion, frozen and object-grounded representation probes,
021 and object-centric decomposition; none of the tested
022 configurations reliably closes the gap, and a label-
023 sensitivity analysis (an automated audit with an inde-
024 pendent human re-audit; agreement in the supplemen-
025 tary) shows that the pattern persists after excluding
026 examples flagged as ambiguous.

027 Accuracy also fails to capture relational coherence.
028 LM-side adaptation significantly improves consistency
029 over zero-shot, while hard-negative training raises
030 facing/facing-away consistency from 66.0% to 77.7%
031 with facing accuracy unchanged at 68.9%; the addi-
032 tional pooled consistency gain over LM-only is not sig-
033 nificant. Cross-dataset evaluation on SITE shows that
034 VSR-trained adaptation does not transfer cleanly: it is
035 neutral overall and significantly harms official spatial-
036 relationship reasoning (75.1%→71.6%, $p = 0.004$). A
037 pre-specified orientation-vocabulary slice, frozen before

SITE evaluation, is analyzed as exploratory evidence
rather than as direct replication of the VSR construct.
The results point to a persistent orientation bottleneck
in which orientation is not robustly decodable by the
tested frozen-feature readouts and relational decisions
are not always propagated coherently, while also show-
ing that task-specific spatial adaptation can transfer
poorly. Each component of the audit—the diagnos-
tic ladder, the label-sensitivity check, the accuracy–
coherence separation, and the cross-dataset transfer
protocol—is specified operationally so that it can be ap-
plied by others to other capabilities, benchmarks, and
model families.

1. Introduction

This paper contributes an evaluation practice as much
as a set of model results. We develop a protocol for
testing, stress-testing, auditing, comparing, and inter-
preting one capability across vision–language models
(VLMs), and instantiate it for object-relative orienta-
tion: facing, facing away, parallel, and perpendicular.
This capability is relevant to robotics, embodied agents,
navigation, and scene understanding, but it is easy to
hide behind an aggregate score. A useful VLM must
identify the relevant objects, infer their relation, and
answer complementary formulations coherently; suc-
cess on one of these steps does not guarantee success
on the others.

Prior benchmarks establish that grounded spatial
reasoning remains difficult, but they leave open how
to diagnose a relation-specific failure. VSR intro-
duced a controlled natural-image probe and reported
particularly poor object-orientation performance [7];
What’sUp and GSR-Bench similarly expose failures on
basic grounded relations [6, 9]; and broader evaluations
continue to identify orientation as a difficult factor [15].
We therefore ask three narrower questions: does the
VSR orientation weakness persist in modern generative

038
039
040
041
042
043
044
045
046
047
048
049
050

051

052
053
054
055
056
057
058
059
060
061
062
063
064

065
066
067
068
069
070
071
072
073
074

075 VLMs and at frontier scale; can prompting, adaptation,
076 or visual readouts remove it; and do accuracy, logi-
077 cal coherence, and cross-dataset transfer tell the same
078 story?

079 In our VSR experiments, the distinction between
080 relation families is striking. Moving from SmolVLM2-
081 2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-
082 creases zero-shot overall accuracy from 74.0% to 80.9%.
083 Horizontal reasoning rises from 70.2% to 84.8%, depth
084 from 68.9% to 75.2%, and containment from 83.4%
085 to 89.3%. Orientation, by contrast, moves only from
086 62.8% to 63.5%. A frontier-scale 2026 model (MiMo-
087 V2.5, 310B total / 15B active parameters) reproduces
088 the same relation-specific pattern: 79.5% overall ac-
089 curacy yet 65.7% on orientation, again inside the
090 62–66% band. We then evaluate interventions span-
091 ning prompting, language-side adaptation, targeted
092 data construction, visual-side adaptation, representa-
093 tion probing, and explicit object-centric decomposition.
094 Structured prompting significantly hurts performance;
095 general and targeted LoRA improve aggregate accuracy
096 but do not remove the orientation ceiling; hard nega-
097 tives improve targeted relational coherence without a
098 significant overall orientation-accuracy gain; and the
099 tested visual-side LoRA configurations do not outper-
100 form the language-side control.

101 As a sensitivity analysis, removing examples flagged
102 by the automated audit raises orientation accuracy
103 (65.7%→78.5% for 7B General LoRA, 66.4%→76.6%
104 for hard-negative LoRA on the strictest 107-example
105 subset) but does not eliminate error (2B structured
106 shows no gain). The exclusion sets are annotator-
107 specific: an independent human re-audit under the
108 same taxonomy reaches only moderate per-item agree-
109 ment ($\kappa=0.36$ on the clean/ambiguous flag) yet repro-
110 duces the same qualitative conclusions; details are in
111 the supplementary. Because the removed examples
112 were drawn from persistent failures, part of this gain is
113 expected by construction.

114 A second failure mode emerges from logical consis-
115 tency. The zero-shot 7B model is consistent on only
116 36.9% of complementary facing/facing-away pairs. LM-
117 side LoRA substantially improves coherence, and hard-
118 negative training raises facing consistency to 77.7%
119 while facing accuracy remains unchanged at 68.9%.
120 The extra hard-negative gain is not significant in the
121 pooled strict-family comparison, yet accuracy stays un-
122 changed while consistency moves: the two metrics cap-
123 ture different behavior.

124 Finally, we test cross-dataset transfer. The 7B
125 model is strong on SITE’s official spatial-relationship-
126 reasoning category (75.1% raw; 59.2% chance-adjusted
127 accuracy), yet the VSR-trained adapter does not

generalize cleanly: it is statistically neutral overall
(54.2%→53.8%, $p = 0.66$) and significantly degrades
the official category (75.1%→71.6%, $p = 0.004$). A
pre-specified, non-official orientation-vocabulary slice
is much weaker in the aggregate (48.3% raw; 24.0%
chance-adjusted) and shows no positive transfer, but—
as we detail below—its composition limits what it can
establish, and we therefore do not treat SITE as inde-
pendent confirmation of the VSR orientation construct.

Our contributions are four reusable components of
an evaluation protocol:

- A fine-grained diagnostic ladder. Across ten VSR
conditions, two differently sized models, and a
frontier-scale zero-shot model, several spatial fam-
ilies improve substantially while object-relative ori-
entation remains unusually resistant. The ladder—
scaling, prompting, language-side adaptation, tar-
geted data construction, visual-side adaptation, rep-
resentation probing, and explicit decomposition—is
defined operationally so it transfers to other bench-
marks and relation families.
- A robustness and sensitivity audit. A label-
sensitivity analysis (automated labels with an inde-
pendent human re-audit), frozen/object-grounded
probes, visual-side adaptation, and object-centric
decomposition constrain purely annotation-based or
single-component explanations. The audit practice
(flagged-example exclusion sets, decodability checks
against majority baselines) applies to any bench-
mark whose labels or examples may be ambiguous.
- An accuracy–coherence separation protocol.
Complementary-relation tests on VSR’s strict
relation pairs show that, for the tested conditions,
relational consistency can change substantially
while task accuracy remains fixed. The protocol
measures consistency against verdict-pattern base
rates, so it can be applied wherever complementary
formulations of the same relation exist.
- A cross-dataset transfer protocol. SITE shows
strong official spatial-relationship performance,
while VSR-trained LoRA transfers poorly and sig-
nificantly harms the official SITE category; the
orientation-vocabulary slice is analyzed as ex-
ploratory evidence. The protocol (pre-specified
subsets, chance-adjusted accuracy, paired transfer
tests) is model- and benchmark-agnostic and is re-
ported with negative as well as positive transfer.

The protocol is deliberately modular: each compo-
nent can be applied independently to other capabilities,
benchmarks, and model families. The supplementary
specifies the prompts, subsets, decision rules, and re-
leased artifacts needed to reproduce the audit.

180	2. Related Work	
181	Benchmarks for grounded spatial reasoning. VSR in-	
182	troduced a controlled natural-image benchmark with	
183	66 spatial relations and identified facing and parallel as	
184	difficult relations for the VLMs studied at the time [7].	
185	What'sUp controls object identity while varying spa-	
186	tial configuration and reports a substantial human-	
187	model gap [6]; GSR-Bench broadens grounded evalu-	
188	ation across models and scales [9]. SITE extends the	
189	scope to single-image, multi-image, and video settings	
190	and organizes tasks by dimensions such as orientation,	
191	intrinsic/extrinsic relations, and static/dynamic rea-	
192	soning [15]. SpatialScore and Mind the Gap further	
193	show that spatial competence remains fragmented and	
194	should be separated from general detection or semantic	
195	comprehension [10, 16]. We use VSR not to re-establish	
196	that spatial reasoning is hard, but to turn its orienta-	
197	tion signal into a controlled diagnostic target and use	
198	SITE to test transfer.	
199	Diagnosing and improving spatial failures. Several	
200	lines of work move beyond aggregate accuracy.	
201	AdaptVis relates successful VSR reasoning to attention	
202	on relevant object locations [2], while CREG shows that	
203	attribution evidence in Qwen2-VL can reflect image lay-	
204	out rather than the queried relation [11]. When More Is	
205	Less finds that spatial cues, commonsense information,	
206	and reasoning instructions can help or hurt depend-	
207	ing on the model and setup [1]. Other work probes	
208	visual-concept decodability [13], separates perception	
209	from reasoning in spatial planning [17], or supplies ge-	
210	ometric inputs for 3D spatial understanding [3]. These	
211	studies motivate our combination of frozen/readout di-	
212	agnostics with adaptation and object-centric decompo-	
213	sition, but do not provide the same end-to-end audit	
214	of one relation family across accuracy, coherence, label	
215	sensitivity, and transfer.	
216	Taken together, prior work establishes the problem	
217	and offers several useful diagnostic lenses, but leaves a	
218	methodological gap: benchmark accuracy, representa-	
219	tion access, logical consistency, annotation sensitivity,	
220	and transfer are usually reported separately. We close	
221	that gap with a single, preregistered-in-spirit diagnos-	
222	tic ladder applied to the same relation family. Our	
223	claim is consequently not to discover spatial difficulty,	
224	but to make a persistent orientation weakness measur-	
225	able, falsifiable, and reusable as an evaluation protocol.	
226	Adaptation, specialization, and transfer. Parameter-	
227	efficient adaptation such as LoRA [5] can improve a	
228	benchmark without improving the underlying capabil-	
229	ity. We therefore compare general, targeted, hard-	
	negative, projector, and vision-plus-projector updates,	230
	then evaluate the strongest language-side adapter on	231
	an untouched benchmark. This design distinguishes	232
	within-benchmark specialization from transfer: VSR	233
	adaptation is neutral overall on SITE, does not im-	234
	prove the orientation subset, and significantly degrades	235
	SITE's official spatial-relationship category. Our em-	236
	phasis on negative transfer complements prior bench-	237
	mark work by treating transfer failure as an evaluation	238
	result rather than an omitted outcome.	239
	Representation probing and logical consistency.	240
	Probes can test whether a signal is accessible from	241
	frozen representations, but probe capacity can itself	242
	learn the task, requiring control baselines and cautious	243
	interpretation [4]. Consistency-oriented objectives	244
	such as SAGE use geometric and linguistic dualities	245
	to train spatial reasoning [8]. We instead use probing	246
	and complementary-pair consistency as independent	247
	diagnostics: the former asks what the tested readouts	248
	can decode, while the latter asks whether answers	249
	to logically linked statements agree. This distinction	250
	allows accuracy and coherence to diverge without	251
	claiming either diagnostic alone is a complete causal	252
	explanation.	253
	3. Experimental Setup	254
	3.1. Benchmarks	255
	VSR. We use VSR [7] as the primary diagnostic	256
	benchmark. The random-split test set contains 2,195	257
	examples. The benchmark defines 64 relation labels;	258
	61 appear in the evaluated test split. The orientation	259
	family contains facing, facing away from, parallel to,	260
	and perpendicular to, totaling 137 test examples. All	261
	main VSR results use the same held-out test set and	262
	deterministic True/False parsing.	263
	SITE. We use SITE [15] for cross-dataset transfer	264
	evaluation only and do not train on SITE. The image	265
	evaluation contains 2,591 examples: 1,368 single-image	266
	and 1,223 multi-image examples. Our primary SITE	267
	subset is the official spatial-relationship-reasoning cate-	268
	gory ($n = 993$). We additionally report a transparently	269
	defined, non-official keyword-derived orientation subset	270
	($n = 2,024$), analyzed as exploratory evidence rather	271
	than as direct replication of the VSR construct. SITE	272
	results use the benchmark's native multiple-choice for-	273
	mat and chance-adjusted accuracy (CAA) in addition	274
	to raw accuracy.	275

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger model is Qwen2-VL-7B-Instruct [14]. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [5] is used for all parameter-efficient adaptation conditions. We additionally evaluate a frontier-scale zero-shot model, MiMo-V2.5 [12] (310B total / 15B activated parameters, dedicated 729M-parameter vision encoder, 2026), served through its public API with the identical frozen prompt, thinking disabled, and greedy decoding (App. A). Its role is external validation of the diagnosed pattern; it is not fine-tuned.

General LoRA uses a proportional sample across relation families. Targeted LoRA allocates most training examples to weak orientation/depth/horizontal families with rehearsal from the remaining relations. Hard-negative LoRA introduces audited complementary orientation examples, primarily facing/facing-away inversions. Vision-side conditions adapt either the multi-modal projector or upper vision blocks together with the projector. The frozen experimental snapshot records the final result set.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band. The

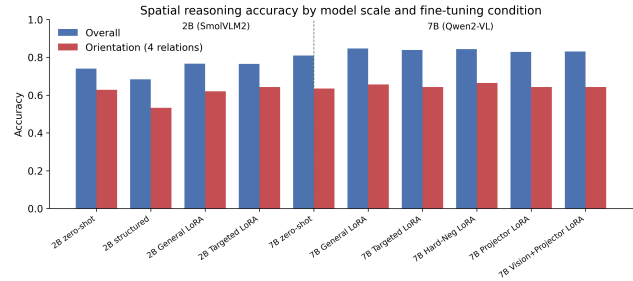


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

frontier-scale zero-shot model (MiMo-V2.5, evaluated identically, App. A) falls in the same band: 79.5% overall and 65.7% orientation, with the orientation family again its weakest relation family (per-family table in the supplementary). The relation-specific pattern is therefore not an artifact of the 2B/7B scale range.

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of information injection on VSR, where additional spatial cues or reasoning instructions can help or hurt depending on the model and setup [1], this result is specific to the tested 2B condition and prompt; it does not establish that prompting generally fails.

4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; the frozen v1 table) identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these values should not be read as a matched comparison against other relation families. A second, independent human re-audit under the same taxonomy (137-example binary flag plus the eight-class taxonomy on the 48 persistent failures) reaches only moderate per-item agreement with the automated labels: on the 48 persistent-failure cases rated by both sources (the automated audit never flagged the remaining 89 orientation examples), $\kappa=0.36$ (66.7% agreement) on the clean/ambiguous flag and Krippendorff's $\alpha=0.14$ on the taxonomy. Yet the human re-audit reproduces the same qualitative conclusions on its strict 107-example subset: orientation accuracy rises (best 77.6%), error is

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6

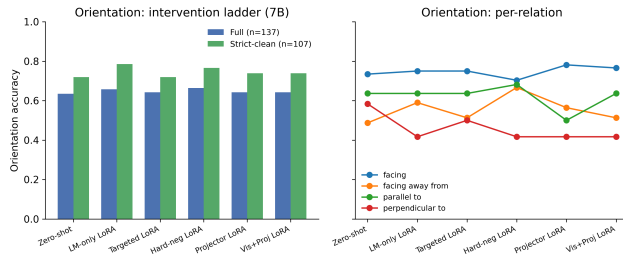


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

not eliminated (the human-audit ceiling of 77.6% is below the automated audit’s 78.5%), and 2B structured prompting still shows no gain (51.4%). The exclusion sets are therefore annotator-specific; the frozen table is retained as v1 and the human re-audit table is reported alongside it in the supplementary. The robust conclusion is narrower: removing flagged examples raises orientation scores but does not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations without fine-tuning the generative model. Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the confidence interval overlaps the majority baseline. In short:

orientation is not robustly decodable by the tested frozen-feature readouts.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [13], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR’s orientation relation family with pooled, patch-vote, and object-grounded readouts over the exact conditioning features of the generative model, and by pairing decodability with adaptation and consistency interventions rather than treating it as an end in itself. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$; these effects do not survive a blanket multiple-comparison

adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not unlock the ceiling.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts agree with the end-to-end control on only 55.1–58.3% of the 127 boxed orientation statements; in the committed analysis (10 no-box statements scored wrong), verdicts diverge on 61–65 statements, and the control is correct while the pipeline is not in 44–47 of them. The committed pipeline scored agreement with the control rather than ground-truth accuracy, so we report the tested decomposition as failing to reproduce the generative model’s behavior, not as a direct accuracy comparison. Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The frontier-scale zero-shot model does not repair coherence either: it reaches 42.7% facing-pair consistency under the same definition (vs. 36.9% for the 7B zero-shot model; 44.0% on the 100 pairs with parseable verdicts, three excluded for non-conforming outputs as reported in the Limitations), with the same False-defaulting verdict pattern as the smaller models (0 both-True, 56.0% both-False verdicts; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The observed verdict patterns show what drives this: the model affirms either member of a facing pair only 17–20% of the time, a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103); ex-

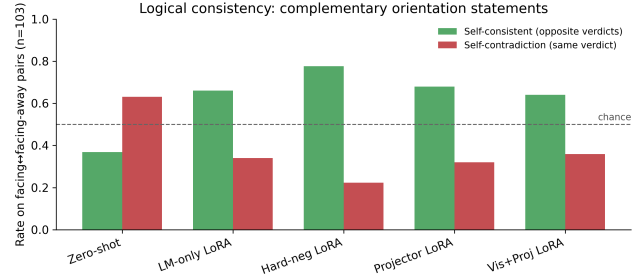


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

pected consistency under verdict independence given these base rates is 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind consistency to 70.7%; pooled strict-family consistency is significantly better than zero-shot ($p < 0.0001$). Hard-negative training further raises facing consistency to 77.7%. Facing accuracy, however, is identical for LM-only and hard-negative LoRA at 68.9%. The additional hard-negative improvement is not significant in the pooled strict-family McNemar comparison ($p = 0.29$). Paired hard-negative supervision therefore provides a targeted intervention on complementary-relation coherence: it descriptively raises facing-pair consistency while leaving original facing accuracy unchanged, although the additional pooled consistency effect over LM-only is not significant. We do not claim a general coherence algorithm; the key result is separability: identical facing accuracy coexists with substantially different facing consistency. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2 (the official category was not broadly weak), this adapter run is a post-protocol cross-dataset transfer check rather than a pre-specified step; the protocol and its outcome are documented in the supplementary. The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer on the independent benchmark; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is substantially weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category, and below the 54.2% / 31.1% all-images average) and shows no positive transfer under the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than as direct replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16], $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in



Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation that orientation relations are difficult [7], while extending it to modern generative VLMs and a much broader intervention analysis.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism, and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official spatial-reasoning subset. Hard negatives improve a targeted coherence behavior while producing relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated automatically with general spatial-reasoning improvement.

8. Limitations

Statistical power is limited at the scale of the orientation diagnostic. The VSR orientation family contains only 137 test examples, and the smallest per-relation subset, perpendicular, has $n=12$; perpendicular results are therefore reported descriptively throughout. The Wilson 95% confidence intervals on the

family-level point estimates are correspondingly wide—62.8% [54, 70], 63.5% [55, 71], 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA ($n=137$ each)—so the trained conditions’ 62–66% band should be read as a set of statistically overlapping point estimates rather than a fine-grained ranking. Several smaller point estimates have confidence intervals that include the majority baseline or chance: the patch-vote parallel/perpendicular probe (73.5%, $n=34$; 95% CI [57, 85] vs. the 64.7% majority), the perpendicular subset for the trained 7B conditions (41.7%, $n=12$; [19, 68] vs. 50%), and the 7B zero-shot facing-away subset (48.7%, $n=39$; [34, 64]). Paired McNemar tests on the orientation family rest on small discordant-pair counts and have limited power: the non-significant projector and vision+projector orientation comparisons ($p=0.86$, $p=0.85$) indicate an absence of detectable effect, not evidence of equivalence. Where the paper draws conclusions, it relies on the full-set paired tests and the Holm-adjusted battery in the supplementary rather than on these small- n point estimates alone.

The clean-label audit’s exclusion sets originate from an automated LLM audit: an independent human re-audit reached only moderate per-item agreement ($\kappa=0.36$ on the clean/ambiguous flag; Krippendorff’s $\alpha=0.14$ on the eight-class taxonomy) yet reproduced the qualitative conclusions, so the sets are annotator-specific, and the frozen table is reported as v1 with the human re-audit table alongside. The SITE orientation slice is keyword-derived rather than an official benchmark category; a composition-confound analysis shows its aggregate score is largely explained by task composition, so we do not treat SITE as independent confirmation of the VSR orientation bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full end-to-end retraining. We study two differently sized VLMs from different families plus a frontier-scale third model; this is a cross-model comparison, not a controlled scale experiment. All trained conditions are single-checkpoint runs: the paired McNemar tests quantify disagreement on the fixed test examples, not training-run variability, and these single-seed results do not estimate it. The MiMo-V2.5 evaluation is served through a public API: 3.1% of its outputs did not conform to the one-word format and are counted as wrong (invalid outputs are never guessed), and 294 responses contained chain-of-thought text alongside the verdict; both are disclosed in the committed predictions (App. A). The MiMo-V2.5 row was evaluated on 2026-08-12 through the OpenCode Go

gateway; models served through public APIs can be updated by the provider without notice, so reproducibility of this specific row depends on the serving snapshot (evaluation date, model id, and per-request token accounting are recorded in results/mimo/). The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results, so lack of improvement should be interpreted for the tested configurations rather than as impossibility claims about all prompting, adaptation, or architectural methods.

9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the VLMs studied here. A larger VLM and standard adaptation substantially improve several spatial families while leaving object-relative orientation comparatively resistant, and a frontier-scale 2026 model reproduces the same relation-specific pattern. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations, while complementary-relation tests show that accuracy and coherence are distinct axes of spatial competence that both remain weak in the frontier model. Cross-dataset evaluation on SITE further shows that VSR-specific adaptation transfers poorly and can negatively transfer to official spatial-relationship reasoning. The results motivate spatial evaluation that reports relation-specific performance, transfer, and logical coherence rather than relying on aggregate accuracy alone.

References

- [1] Muku Akasaka and Soyeon Caren Han. When more is less: A systematic analysis of spatial and commonsense information for visual spatial reasoning. arXiv preprint arXiv:2602.21619, 2026. 3, 4
- [2] Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. In Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025. 3
- [3] Erik Daxberger, Nina Wenzel, David Griffiths, Haiming Gang, Justin Lazarow, Gefen Kohavi, Kai Kang, Marcin Eichner, Yinfei Yang, Afshin Dehghan, and Peter Grasch. MM-Spatial: Exploring 3d spatial understanding in multimodal LLMs. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 7395–7408, 2025. 3
- [4] John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Con-

- ference on Natural Language Processing, pages 2733–2743. Association for Computational Linguistics, 2019. 3, 5
- [5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. arXiv preprint arXiv:2106.09685, 2021. 3, 4
- [6] Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? investigating their struggle with spatial reasoning. arXiv preprint arXiv:2310.19785, 2023. 1, 3
- [7] Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. Transactions of the Association for Computational Linguistics, 11:635–651, 2023. 1, 3, 7
- [8] Junming Liu, Yuqi Li, Yifei Sun, Maonan Wang, Piotr Koniusz, Yirong Chen, and Ding Wang. Self-evolving spatial reasoning in vision language models via geometric logic consistency. arXiv preprint arXiv:2605.18162, 2026. 3, 6
- [9] Navid Rajabi and Jana Kosecka. GSR-BENCH: A benchmark for grounded spatial reasoning evaluation via multimodal LLMs. arXiv preprint arXiv:2406.13246, 2024. 1, 3
- [10] Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsaftaris. Mind the gap: Benchmarking spatial reasoning in vision-language models. arXiv preprint arXiv:2503.19707, 2025. 3
- [11] Kaizhen Tan, Yang Feng, and Heqing Du. CREG: Compass relational evidence graph for characterizing directional structure in VLM spatial-reasoning attribution. arXiv preprint arXiv:2603.20475, 2026. 3, 5
- [12] Xiaomi MiMo Team. Mimo-v2.5. <https://huggingface.co/collections/XiaomiMiMo/mimo-v25>, 2026. 4
- [13] Nikos Theodoridis, Reenu Mohandas, Ganesh Sistu, Anthony Scanlan, Ciarán Eising, and Tim Brophy. Probing visual concepts in lightweight vision-language models for automated driving. Transactions on Machine Learning Research, 2026. 3, 5
- [14] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-VL: Enhancing vision-language model’s perception of the world at any resolution. arXiv preprint arXiv:2409.12191, 2024. 4
- [15] Wenqi Wang, Reuben Tan, Pengyue Zhu, Jianwei Yang, Zhengyuan Yang, Lijuan Wang, Andrey Kolobov, Jianfeng Gao, and Boqing Gong. SITE: towards spatial intelligence thorough evaluation. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 9058–9069, 2025. 1, 3, 6
- [16] Haoning Wu, Xiao Huang, Yaohui Chen, Ya Zhang, Yanfeng Wang, and Weidi Xie. Spatialscore: Towards comprehensive evaluation for spatial intelligence. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 31029–31041, 2026. 3
- [17] Qiucheng Wu, Handong Zhao, Michael Saxon, Trung Bui, William Yang Wang, Yang Zhang, and Shiyu Chang. VSP: Diagnosing the dual challenges of perception and reasoning in spatial planning tasks for MLLMs. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 2270–2280, 2025. 3